

Lizzy Grynova
3/31/2026
EMEC 103 Section 001
Adam Michalson

EMEC103 Project 1 - Mousetrap Car

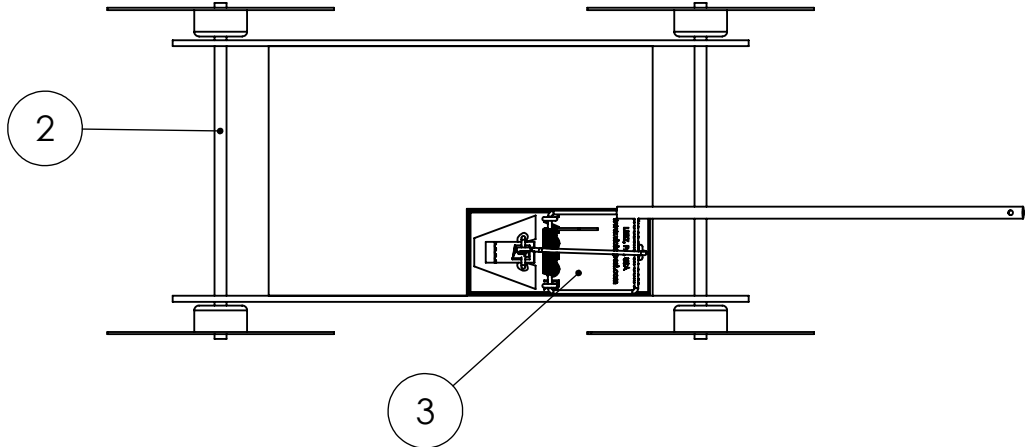
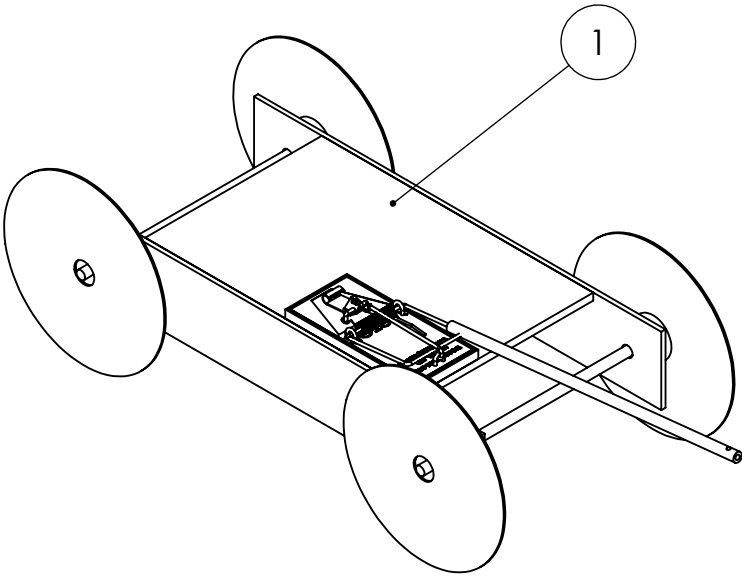
The following items are included in this submission

- 1.) Preliminary Design Sketch
- 2.) Preliminary Design Drawings
- 3.) Detailed Work Instructions
- 4.) Prototype
- 5.) Peer Presentation

I **pledge** on my **honor** that the work included here is my own original work. I have not given nor received any unauthorized assistance on this assignment.

Signed Lizzy Grynova

ITEM NO.	PART NUMBER	QTY.
1	Body & MT Assembly	1
2	Axle Subassembly	2
3	Mousetrap Assembly	1



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± 1/16" ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL ± 0.03" THREE PLACE DECIMAL ± 0.005"		NAME	DATE	MSU M.& I.E. DEPT.		
		DRAWN	LG	4/1	TITLE: Mouse Trap Car		
		CHECKED					
		INTERPRET GEOMETRIC TOLERANCING PER:	ENG APPR.				
	COMMENTS:	MFG APPR.			SIZE	DWG. NO.	REV
					A 103-ASM-1		
NEXT ASSY	DO NOT SCALE DRAWING	MATERIAL VARS					
					SCALE: 1:4 WEIGHT: 0.61 SHEET 1 OF 1		

Manufacturing Plan

Part Name: Mousetrap Car Assembly

Material Body & MT Assembly, Axle Subassemblies

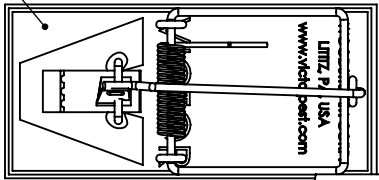
Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive mousetrap assembly	N/A	N/A
20	Receive body and attach side panels and mousetrap assembly	N/A	Hot glue gun, ruler
30	Insert axle subassemblies through side panel holes	N/A	~15" string
40	Tie string through hole in arm extension and secure with knot	N/A	N/A
50	Attach string loop to zip tie on axle and wind string around axle	N/A	N/A

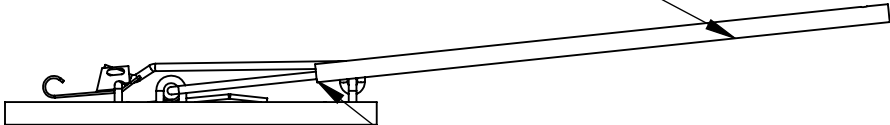
ITEM NO.	PART NUMBER	QTY.
1	Arm Extension	1
2	Mousetrap	1

2

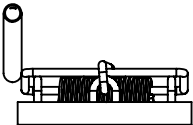


Pass string through the drilled hole, exit through the rod opening, and secure with a knot to create a loop.

1



Secure arm extension to bar using multiple zip ties; position as far forward as possible while still being secure



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± 1/16" ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL ± 0.03" THREE PLACE DECIMAL ± 0.005"			NAME	DATE	MSU M.& I.E. DEPT.			
		DRAWN	LG	3/3	TITLE: Mousetrap Assembly			
		CHECKED						
		ENG APPR.						
INTERPRET GEOMETRIC TOLERANCING PER:		MFG APPR.			SIZE	DWG. NO.	REV	
COMMENTS:		MATERIAL VARS			A	103-ASM-2		
NEXT ASSY	DO NOT SCALE DRAWING				SCALE: 1:2		WEIGHT: 0.04	SHEET 1 OF 1

Manufacturing Plan

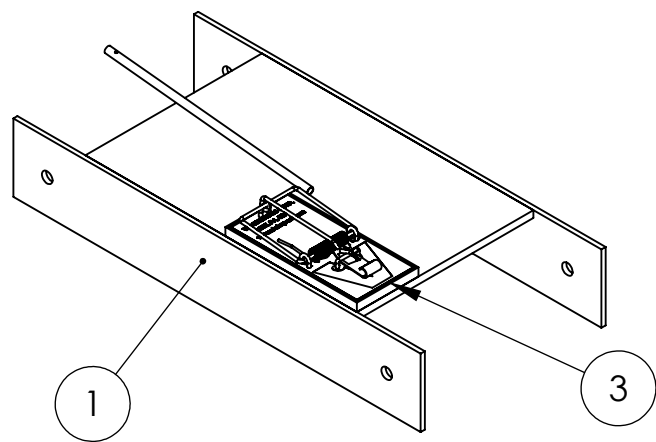
Part Name: Mousetrap Assembly.

Material: Mousetrap & Arm Extension

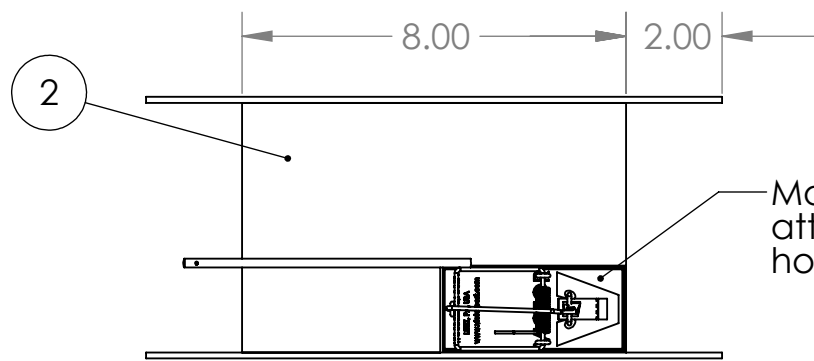
Developed by: Lizzy Grynova

Date: 3/26/2026

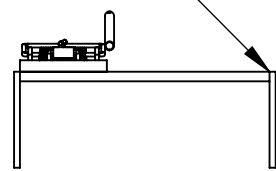
OP #	Operation	Machine(s)	Tooling
10	Receive mousetrap and arm extension	N/A	N/A
20	Attach arm extension to mousetrap arm with multiple zipties	N/A	Zip ties
30	Ship to Assembly	N/A	N/A



ITEM NO.	PART NUMBER	QTY.
1	Side Panel	2
2	Body	1
3	Mousetrap Assembly	1



Side Panels will be attached to the body using hotglue



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm 1/16"$ ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL $\pm 0.03"$ THREE PLACE DECIMAL $\pm 0.005"$ INTERPRET GEOMETRIC TOLERANCING PER:		NAME	DATE	MSU M.& I.E. DEPT.	
		DRAWN	LG		
		CHECKED		TITLE: Body & MT Assembly	
		ENG APPR.			
		MFG APPR.		SIZE	DWG. NO.
COMMENTS:		MATERIAL VARS		A	103-ASM-3
NEXT ASSY	DO NOT SCALE DRAWING			SCALE: 1:2	WEIGHT: 0.18
				SHEET 1 OF 1	

Manufacturing Plan

Part Name: Body and MT Assembly

Material: Body, Side panels, Mousetrap Assembly

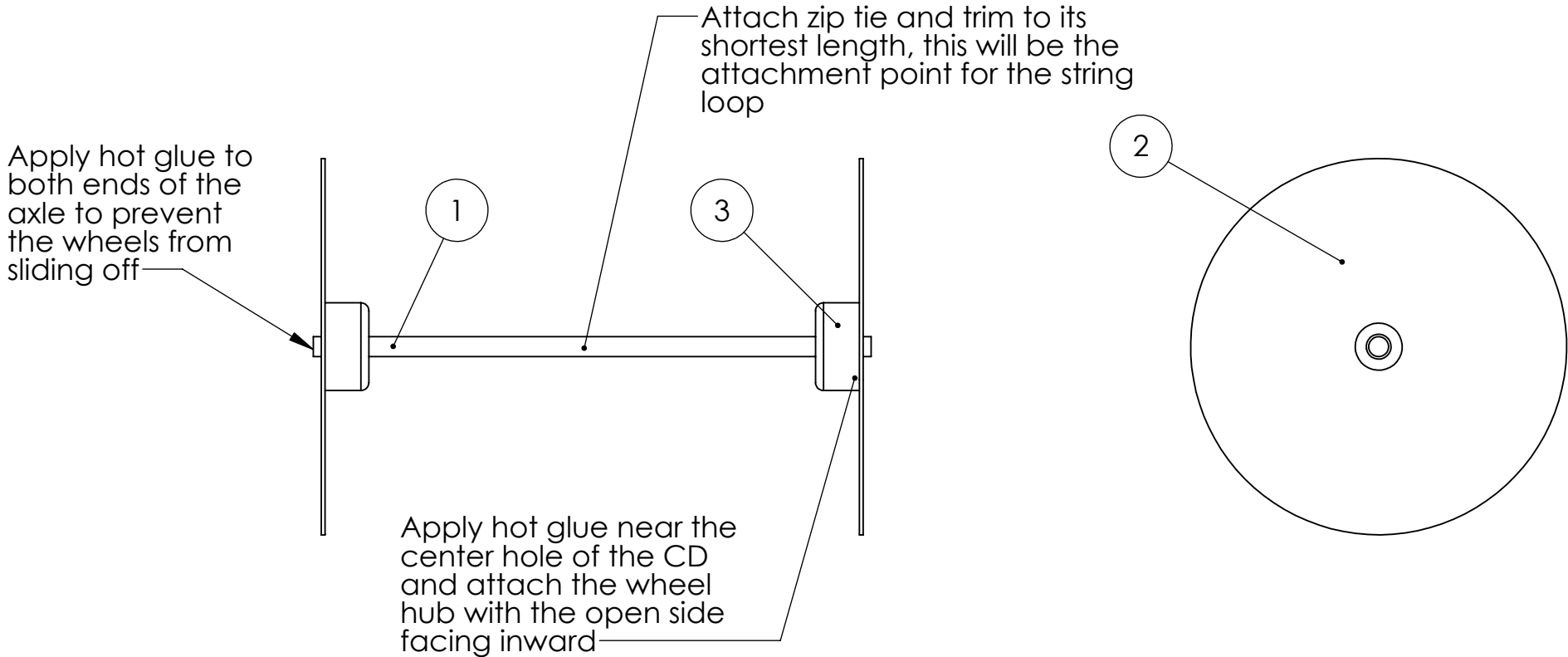
Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive body, side panels, and mousetrap assembly	N/A	N/A
20	Measure and mark side panel locations on body	N/A	Ruler, Pencil
30	Attach side panels to body using hot glue, ensure panels are parallel	N/A	Hot glue gun
40	Attach mousetrap assembly to body using hot glue	N/A	Hot glue gun

IMPORTANT: Do not assemble before putting through the side panels

ITEM NO.	PART NUMBER	QTY.
1	Axle	1
2	CD Wheel	2
3	Wheel Hub	2
4	Ziptie	1



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm 1/16$ " ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL ± 0.03 " THREE PLACE DECIMAL ± 0.005 " INTERPRET GEOMETRIC TOLERANCING PER:		NAME	DATE	MSU M.& I.E. DEPT.	
		DRAWN	LG	4/1	TITLE: Axle Subassembly
		CHECKED			
		ENG APPR.			SIZE DWG. NO. REV
COMMENTS:		MFG APPR.			A 103-SUB-1
NEXT ASSY		DO NOT SCALE DRAWING		SCALE: 1:2 WEIGHT: 0.18 SHEET 1 OF 1	

Manufacturing Plan

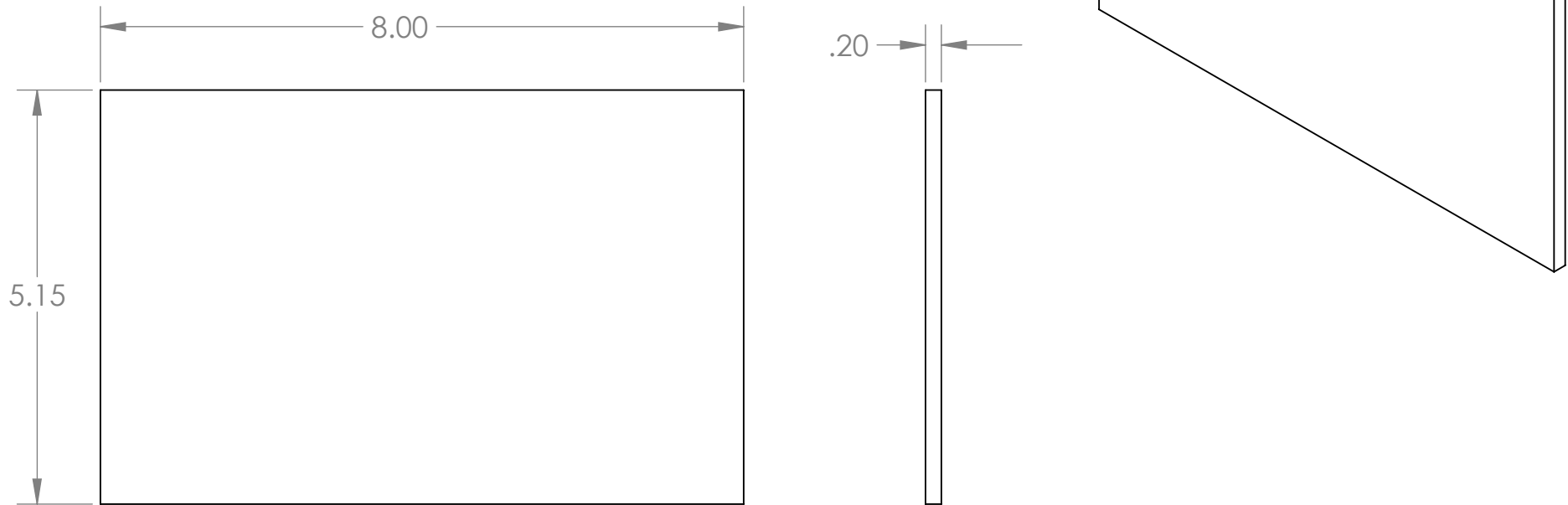
Part Name: Axle Subassembly

Material Steel, CD wheels, wheel hubs, and zip tie

Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive axle, CD wheels, wheel hubs, and zip tie	N/A	N/A
20	Attach wheel hubs to CD wheels using hot glue (open side inward)	N/A	Hot glue gun
30	Install wheel and hub assemblies onto axle	N/A	N/A
40	Attach zip tie to axle and trim to shortest length	N/A	Zip tie, cutters
50	Apply hot glue to both ends of axle to prevent wheels from sliding off	N/A	Hot glue gun



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL $\pm 1/16"$
ANGULAR: MACH $\pm .5$ BEND ± 1
TWO PLACE DECIMAL $\pm 0.03"$
THREE PLACE DECIMAL $\pm 0.005"$

INTERPRET GEOMETRIC
TOLERANCING PER:

COMMENTS:

DO NOT SCALE DRAWING

NAME

DATE

MSU M.& I.E. DEPT.

TITLE:

Body

SIZE

A

DWG. NO.

103-P1-001

REV

SCALE: 1:2

WEIGHT: 0.05

SHEET 1 OF 1

NEXT ASSY

MATERIAL

Balsa

Manufacturing Plan

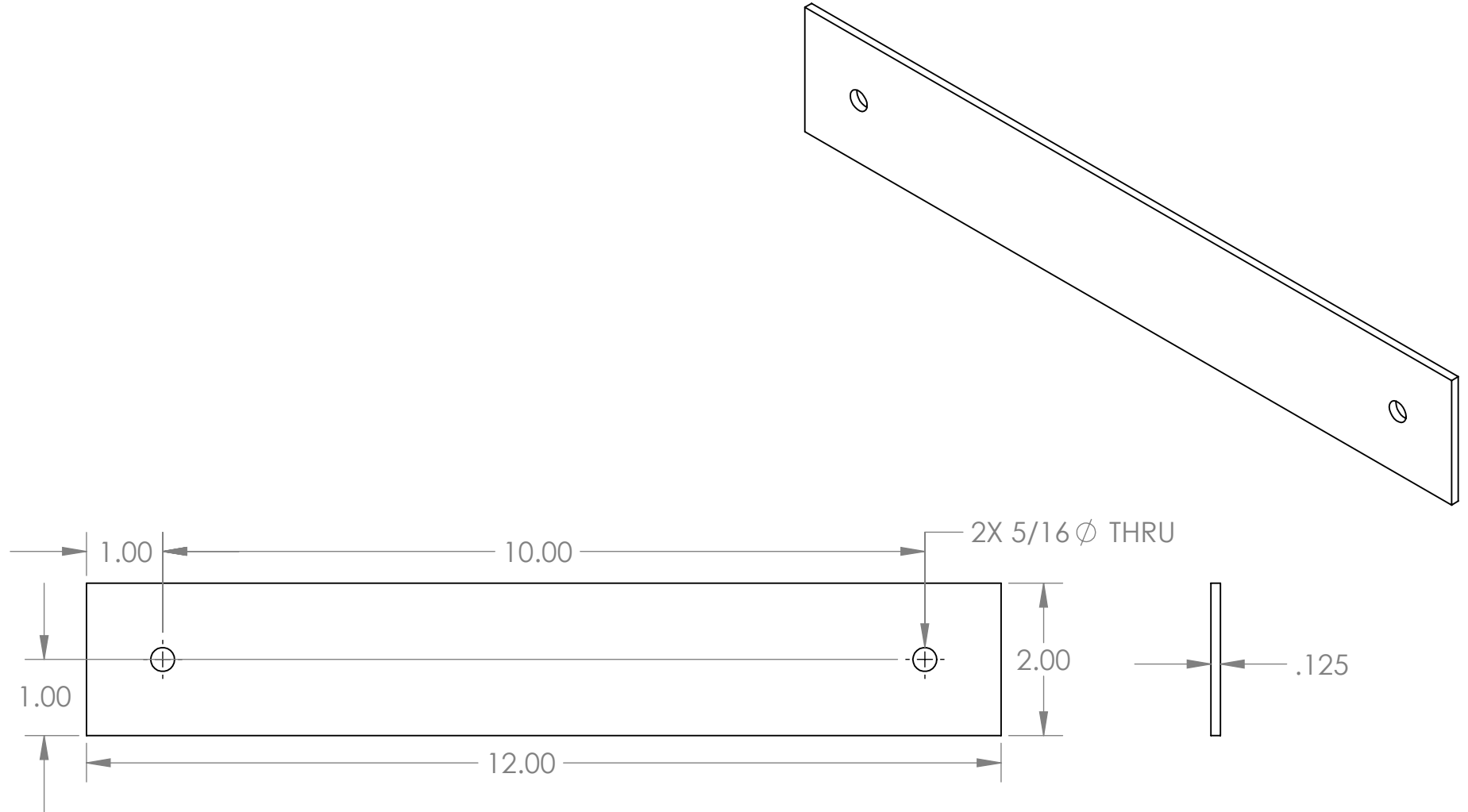
Part Name: Body

Material: Balsa Wood

Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive stock	N/A	N/A
20	Mark out shape and slot location	N/A	Pencil, Ruler
30	Cut along marked lines	N/A	Hand saw and clamp
40	Sand edges and verify dimensions	N/A	Sandpaper
50	Ship to assembly	N/A	N/A



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

		UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm 1/16"$ ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL $\pm 0.03"$ THREE PLACE DECIMAL $\pm 0.005"$		NAME	DATE	MSU M.& I.E. DEPT.	
		INTERPRET GEOMETRIC TOLERANCING PER:		DRAWN	LG	3/26	TITLE: Side Panels
		COMMENTS:		CHECKED			
				ENG APPR.			SIZE DWG. NO. REV
NEXT ASSY		DO NOT SCALE DRAWING		MFG APPR.			A 103-P1-002
				MATERIAL Balsa Wood			
				-	SCALE: 1:2	WEIGHT: 0.02	SHEET 1 OF 1

Manufacturing Plan

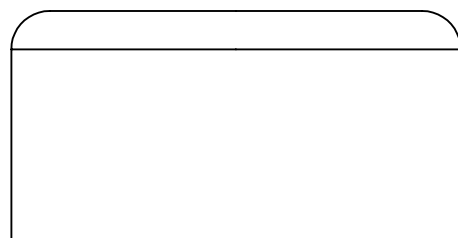
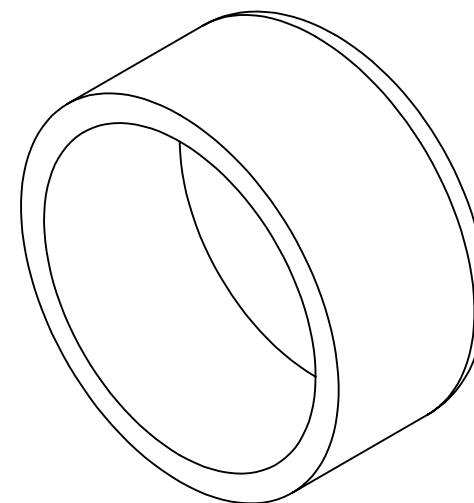
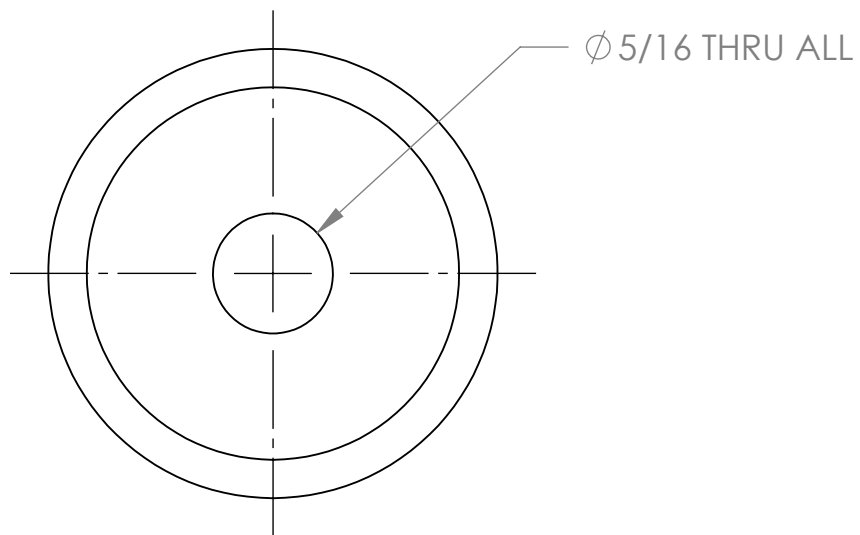
Part Name: Side Panels

Material: Balsa Wood

Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive stock	N/A	N/A
20	Mark out dimensions and hole locations	N/A	Pencil, Ruler
30	Cut panels to size	N/A	Hand saw and clamp
40	Drill axle holes	Hand Drill	5/16" Drill Bit
50	Ship to assembly	N/A	N/A



Made from Bottle Caps

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES		NAME	DATE	MSU M.& I.E. DEPT.	
	TOLERANCES:		DRAWN	LG		
	FRACTIONAL $\pm 1/16"$		CHECKED		TITLE: Wheel Hub	
	ANGULAR: MACH ± 5 BEND ± 1		ENG APPR.			
	TWO PLACE DECIMAL $\pm 0.03"$		MFG APPR.		SIZE	DWG. NO.
	THREE PLACE DECIMAL $\pm 0.005"$				A	103-P1-003
	INTERPRET GEOMETRIC TOLERANCING PER:		MATERIAL			REV
	COMMENTS:		PP Homopolymer			
			Bottle Cap			
NEXT ASSY	DO NOT SCALE DRAWING		SCALE: 2:1		WEIGHT: 0.01	SHEET 1 OF 1

Manufacturing Plan

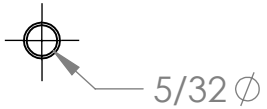
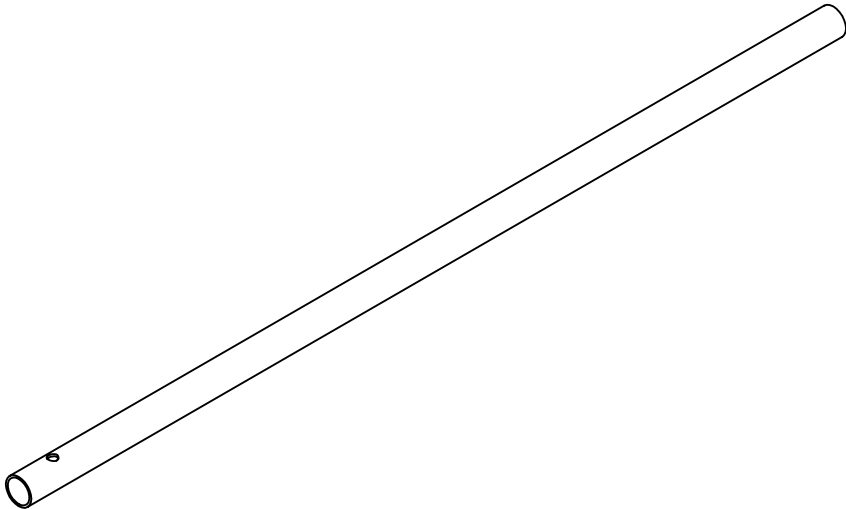
Part Name: Wheel Hub

Material: Bottle Cap

Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive bottle cap	N/A	N/A
20	Mark center point	N/A	Sharpie
30	Drill hole	Hand Drill	5/16" Drill Bit
40	Ship to assembly	N/A	N/A



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

NEXT ASSY	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm 1/16$ " ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL ± 0.03 " THREE PLACE DECIMAL ± 0.005 "	DRAWN	LG	3/26	MSU M.& I.E. DEPT.		
	INTERPRET GEOMETRIC TOLERANCING PER:	CHECKED			TITLE: Arm Extension		
	COMMENTS:	ENG APPR.			SIZE A	DWG. NO. 103-P1-004	REV
		MFG APPR.					
		MATERIAL 3/16 Brass Rod			SCALE: 1:1	WEIGHT: 0.01	SHEET 1 OF 1

Manufacturing Plan

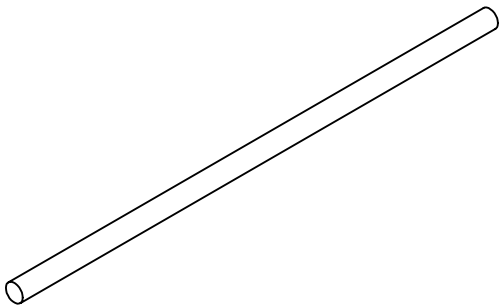
Part Name: Arm Extension

Material: Brass Tube

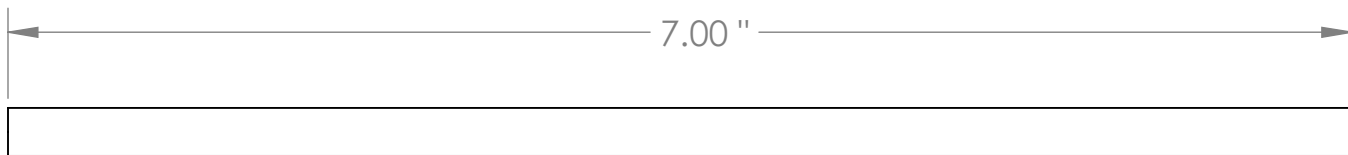
Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive tube stock	N/A	N/A
20	Measure and mark hole location	N/A	Sharpie, Ruler
30	Drill hole	Hand Drill	1/16" Drill Bit
40	Ship to assembly	N/A	N/A



Ø .25



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

	UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± 1/16" ANGULAR: MACH ± 5 BEND ± 1 TWO PLACE DECIMAL ± 0.03" THREE PLACE DECIMAL ± 0.005"		NAME	DATE	MSU M.& I.E. DEPT.			
		DRAWN	LG	3/23	TITLE: Axle			
		CHECKED						
		INTERPRET GEOMETRIC TOLERANCING PER:	ENG APPR.					
	COMMENTS:	MFG APPR.			SIZE	DWG. NO.	REV	
					A	103-P1-005		
NEXT ASSY	DO NOT SCALE DRAWING	MATERIAL Steel			SCALE: 1:1		WEIGHT: 0.11	SHEET 1 OF 1

Manufacturing Plan

Part Name: Axle

Material: Steel Rod

Developed by: Lizzy Grynova

Date: 3/26/2026

OP #	Operation	Machine(s)	Tooling
10	Receive rod stock	N/A	N/A
20	Measure and mark length	N/A	Sharpie, Ruler
30	Cut to length	N/A	Hacksaw, Clamp
40	Smooth edges	N/A	Sandpaper
50	Ship to assembly	N/A	N/A